

CYBERSNOVA CYBER SECURITY INTERNSHIP

FINAL PROJECT REPORT

Domain: Cyber Security

Levels Completed: Beginner + Advanced

Beginner: Projects 1–4

Advanced: OverTheWire Bandit Levels 0–20

Submitted By: VARSHA KAMMARI

Registration Number: CSN-SP-2026-251

Internship Period: SEPTEMBER–2026

Organization: CyberSNova

Submission Date: 26 September

DECLARATION

I hereby declare that this report documents the cybersecurity practical work completed by me as part of the CyberSNova Cyber Security Internship.

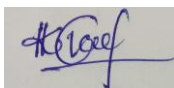
The practical activities were performed using the authorized training targets and platforms provided for the internship. The report contains my practical observations, commands used, methodology, screenshots, and learning outcomes.

I understand that cybersecurity techniques should only be used against systems for which appropriate authorization has been provided.

Name: VARSHA KAMMARI

Registration Number: CSN-SP-2026-251

Signature:

A handwritten signature in blue ink, appearing to read 'Varsha Kammari', is shown next to the 'Signature:' label.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to CyberSNova for providing me with the opportunity to complete this Cyber Security Internship.

This internship helped me gain practical exposure to network reconnaissance, service enumeration, directory enumeration, HTTP traffic analysis, web vulnerability assessment, and Linux-based cybersecurity challenges.

I am also thankful for the authorized vulnerable environments and cybersecurity tools that helped me understand the concepts through practical exercises.

ABSTRACT

This report presents the practical work completed during the Cyber Security Internship at CyberSNova. The internship provided hands-on exposure to fundamental cybersecurity concepts, including network reconnaissance, service enumeration, web directory enumeration, HTTP traffic analysis, and basic web vulnerability assessment.

The Beginner Level consisted of four practical projects: Network Reconnaissance & Service Enumeration, Hidden Directory & File Enumeration, Login Traffic Analysis, and Basic Web Vulnerability Assessment. These activities involved tools such as Nmap, WhatWeb, Netcat, Gobuster, Burp Suite Community Edition, Wireshark, and Nikto.

In addition to the Beginner Level, the Advanced Level was completed through the OverTheWire Bandit Challenge, covering Levels 0–20. This work provided practical experience with Linux command-line operations, file and directory handling, text processing, SSH authentication, OpenSSL, networking utilities, and privilege-related concepts.

The report documents the objectives, methodologies, commands used, observations, practical evidence, screenshots, and learning outcomes from the completed activities. Overall, the internship strengthened practical understanding of cybersecurity tools, Linux-based security operations, network analysis, and basic web security assessment techniques.

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1. INTRODUCTION

Cybersecurity involves protecting computer systems, networks, applications, and information from unauthorized access and security threats.

Practical security testing helps develop an understanding of how systems can be discovered, enumerated, analyzed, and assessed for security weaknesses.

Level	Project / Challenge	Status
Beginner	Network Reconnaissance & Service Enumeration	Completed
Beginner	Hidden Directory & File Enumeration	Completed
Beginner	Login Traffic Analysis	Completed
Beginner	Basic Web Vulnerability Assessment	Completed
Advanced	OverTheWire Bandit – Levels 0–20	Completed

Tools Used: Nmap, WhatWeb, Netcat, Gobuster, Burp Suite Community Edition, Wireshark, Nikto, SSH, OpenSSL, and Linux command-line utilities.

2. INTERNSHIP OBJECTIVES

The primary objective of this internship was to develop practical knowledge and hands-on skills in the field of **Cyber Security**. The internship focused on understanding basic security concepts and applying them through practical exercises and security tools. To understand basic cybersecurity assessment methodology.

- To perform network reconnaissance.
- To identify accessible ports and services.
- To perform web directory enumeration.
- To analyze HTTP traffic.
- To understand login request behavior.
- To perform basic web vulnerability assessment.
- To develop Linux command-line skills.
- To understand SSH authentication.
- To practice encoding and decoding.
- To understand file permissions.
- To work with network services.
- To develop cybersecurity documentation skills.

3. TOOLS AND ENVIRONMENT

Operating Environment:

Operating System: Kali Linux

Virtualization: Oracle VirtualBox

Environment: Authorized cybersecurity training environment

Tools Used:

Tool	Purpose
Nmap	Network and service enumeration
WhatWeb	Web technology identification
Netcat	Network connectivity testing
Gobuster	Directory enumeration
Burp Suite	HTTP request/response analysis
Wireshark	Network traffic analysis
Nikto	Web server assessment
SSH	Secure remote connection
OpenSSL	SSL/TLS connection testing
Linux utilities	File/search/encoding/compression tasks

4. BEGINNER LEVEL

PROJECT 1

NETWORK RECONNAISSANCE & SERVICE ENUMERATION

4.1 Objective

The objective of this project was to perform information gathering against an authorized vulnerable web application.

The assessment included:

- Target connectivity verification
- Port scanning
- Service enumeration
- Technology identification
- TCP connectivity testing .

4.2 Target

Target: <http://testphp.vulnweb.com/>

4.3 Tools

- Nmap
- WhatWeb
- Netcat
- Kali Linux

4.4 Methodology

1. Verify target connectivity.
2. Perform a basic Nmap scan.
3. Perform service/version enumeration.
4. Analyze the web application using WhatWeb.

5. Test TCP connectivity using Netcat.
6. Record observations.
7. Capture screenshots as evidence.

4.5 Commands Used

ping testphp.vulnweb.com

nmap testphp.vulnweb.com

nmap -sV testphp.vulnweb.com

whatweb testphp.vulnweb.com

nc testphp.vulnweb.com 80

4.6 Screenshot Evidence

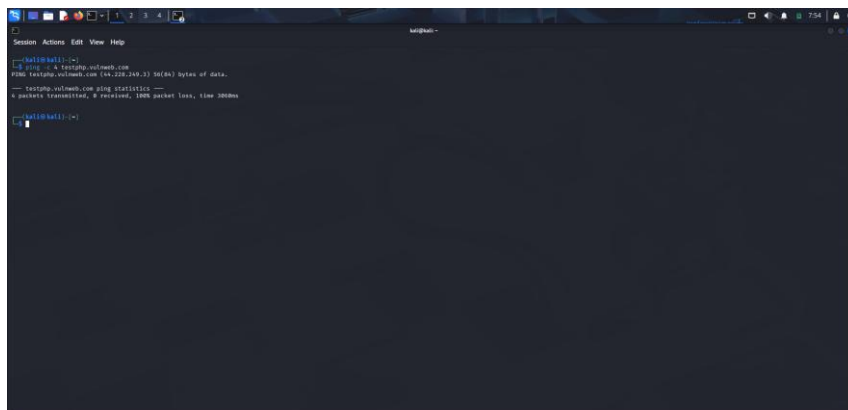


Figure 1: Connectivity Verification

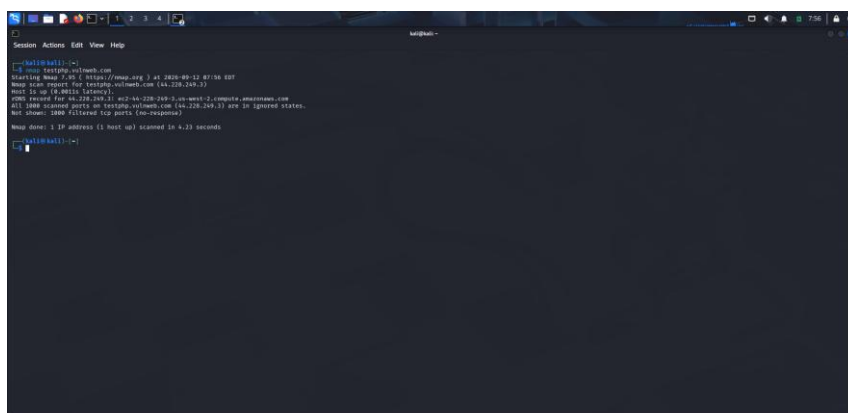


Figure 2: Nmap Basic Scan

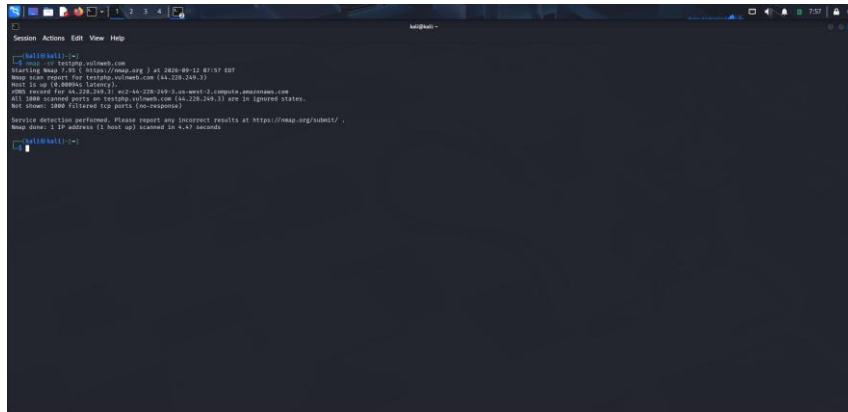


Figure 3: Nmap Service and Version Scan

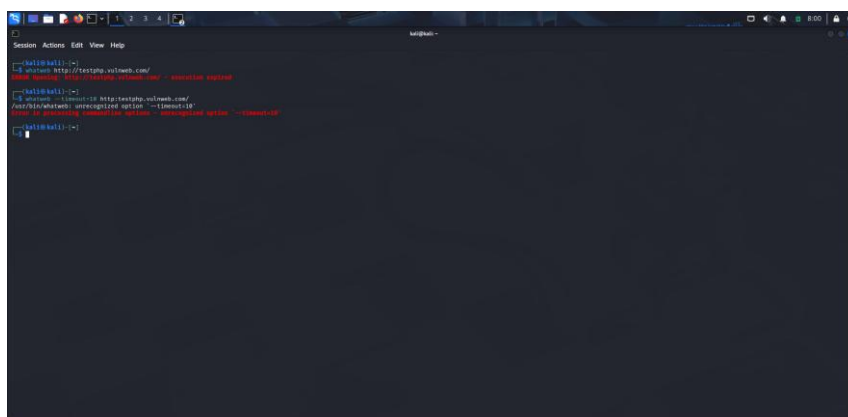


Figure 4: WhatWeb Scan Result

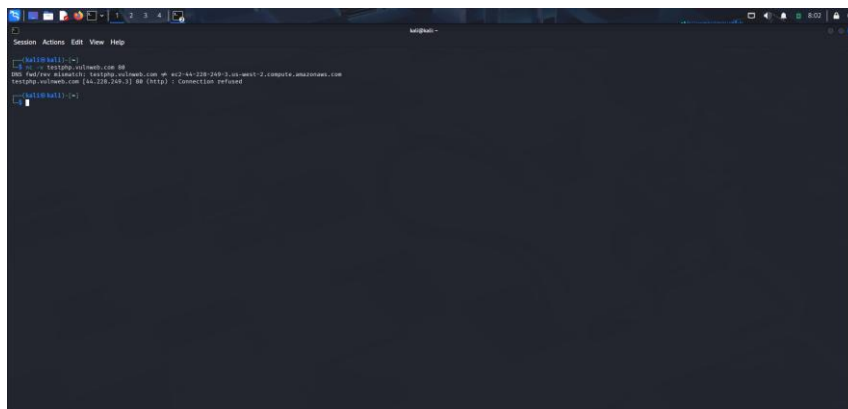


Figure 5: Netcat Port Testing

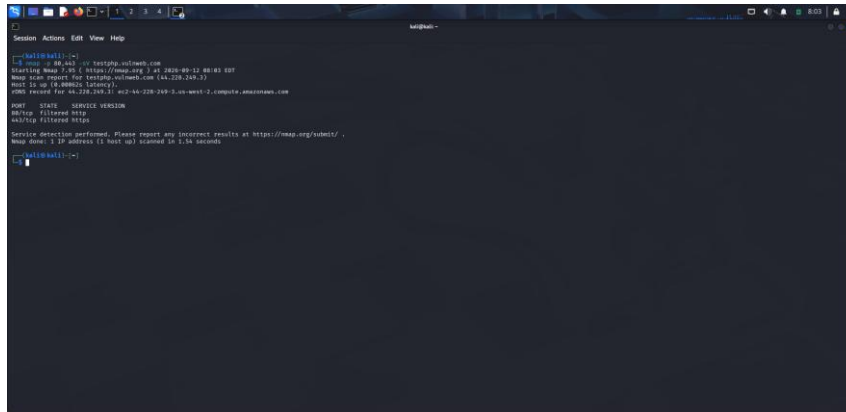


Figure 6: Web Port Identification

4.7 Findings

The reconnaissance activity successfully demonstrated:

- Target connectivity testing.
- Network port scanning.
- Service/version detection.
- Web technology identification.
- TCP connectivity testing.

Important observed ports/services:

Port	State	Service
------	-------	---------

80/tcp	Filtered	HTTP
--------	----------	------

443/tcp	Filtered	HTTPS
---------	----------	-------

4.8 Conclusion

This project helped me understand the initial reconnaissance phase of cybersecurity assessment and provided practical experience with Nmap, WhatWeb, and Netcat.

PROJECT 2

HIDDEN DIRECTORY & FILE ENUMERATION

5.1 Objective

The objective was to identify hidden directories and resources in the authorized vulnerable web application.

CyberSNova specifies Gobuster, Dirsearch, and Dirb as recommended tools for this project.

5.2 Target

`http://testasp.vulnweb.com/`

5.3 Tool Used

- Gobuster
- Kali Linux
- Web browser

5.4 Methodology

1. Verify target connectivity.
2. Verify the HTTP response.
3. Run directory enumeration.
4. Analyze discovered resources.
5. Verify interesting results.
6. Document the findings.

5.5 Command Used

```
gobuster dir -u http://testasp.vulnweb.com/ -w /usr/share/wordlists/dirb/common.txt
```

5.6 Screenshot Evidence

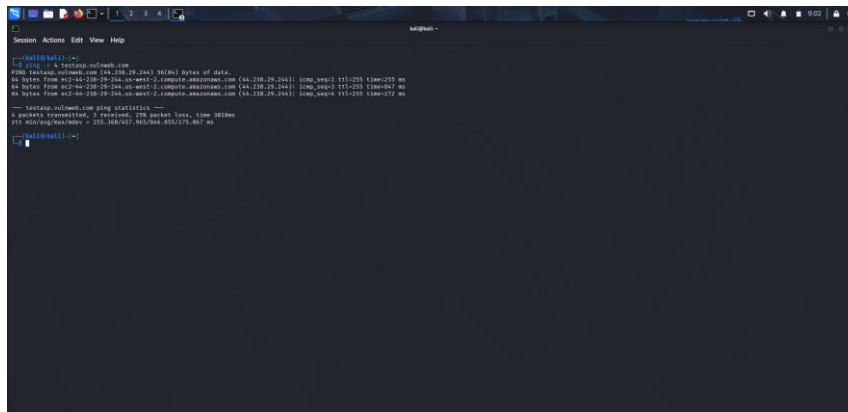


Figure 7: Target Connectivity Verification

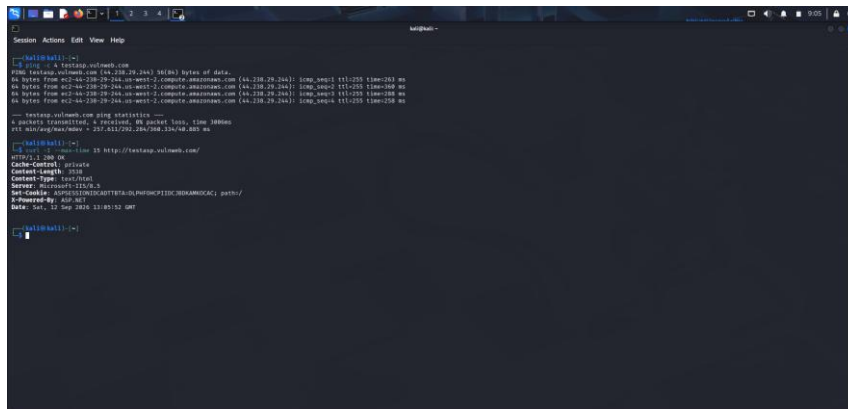


Figure 8: HTTP Response Verification

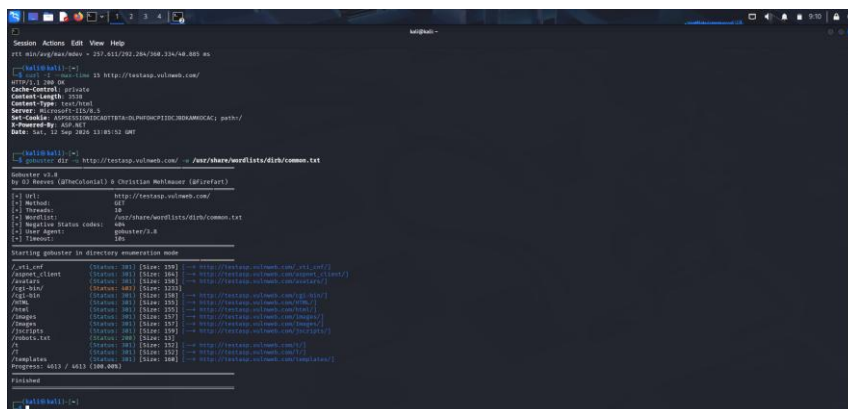


Figure 9: Gobuster Directory Enumeration



No.	Resource	Status	Observation
1	/cgi-bin/	403 Forbidden	Resource exists but access is restricted.
2	/robots.txt	200 OK	Resource is accessible and returned successfully.
3	/avatars/	301 Moved Permanently	Redirects to the trailing-slash URL.

5.8 Security Relevance

Exposed directories and files may reveal application functionality or resources that were not intended to be publicly accessible.

Possible risks include information disclosure and exposure of administrative or backup resources.

5.9 Recommendations

- Remove unnecessary files.
- Restrict sensitive directories.
- Protect administrative interfaces.
- Avoid storing backup files in public web directories.
- Regularly perform directory enumeration checks.

5.10 Conclusion

This project provided practical experience in web directory enumeration and helped me understand how hidden resources can increase the attack surface of a web application.

PROJECT 3

LOGIN TRAFFIC ANALYSIS

6.1 Objective

The objective was to capture and analyze login-related network traffic and determine how sensitive information was transmitted.

CyberSNova specifies Wireshark and Burp Suite Community Edition for this project.

6.2 Target

<http://testasp.vulnweb.com/>

6.3 Tools

- Burp Suite Community Edition
- Wireshark
- Web browser

6.4 Methodology

1. Open the authorized target.
2. Perform the login/request operation.
3. Capture the generated traffic.
4. Inspect the HTTP request.
5. Analyze the request URI and parameters.
6. Document security observations.

6.5 Screenshot Evidence

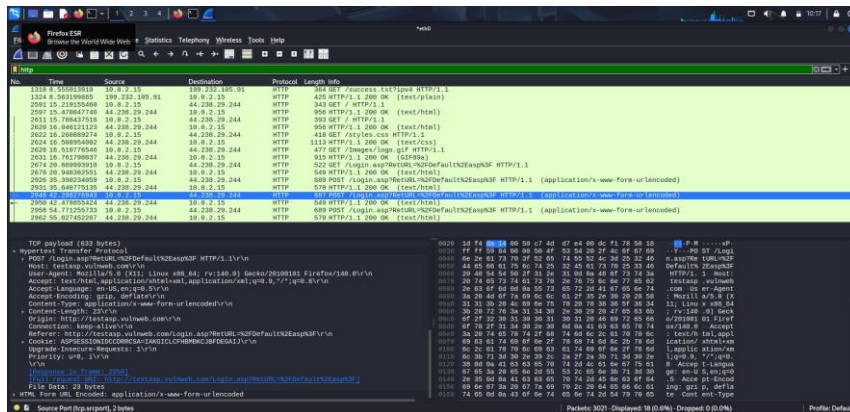


Figure 12: HTTP POST Request URI

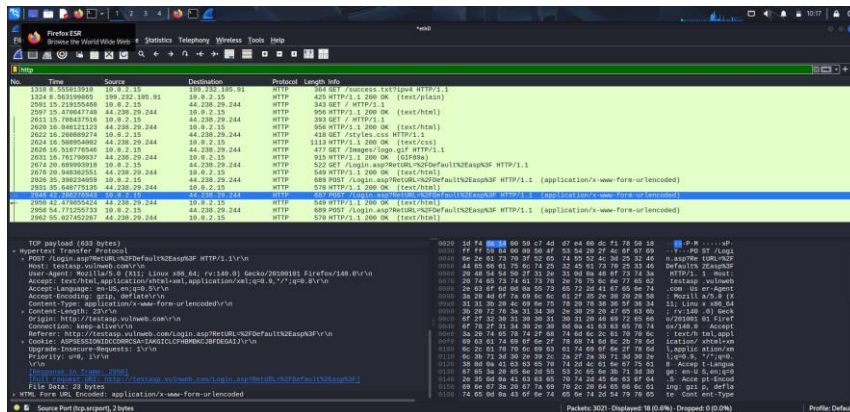


Figure 13: Login Form Data Observed in HTTP POST Request

6.6 Observation

The captured evidence demonstrates an HTTP POST request associated with the login activity.

The traffic analysis focused on:

- HTTP method
- Request URI
- Request parameters
- Transmission protocol
- Visibility of submitted information

6.7 Security Implication

Sensitive authentication information should be transmitted through encrypted HTTPS/TLS communication.

Plain HTTP does not provide encryption for the transmitted application data.

6.8 Recommendations

- Use HTTPS.
- Redirect HTTP to HTTPS.
- Configure TLS correctly.
- Protect authentication information.
- Use secure cookies and appropriate authentication controls.

6.9 Conclusion

This project helped me understand HTTP login traffic and the importance of encrypted communication when handling authentication information.

PROJECT 4

BASIC WEB VULNERABILITY ASSESSMENT

7.1 Objective

The objective was to perform a basic assessment of an authorized vulnerable web application using Burp Suite and Nikto.

CyberSNova requires identification of common web vulnerabilities, severity classification, evidence, and mitigation recommendations.

7.2 Target

<http://testasp.vulnweb.com/>

7.3 Tools

- Burp Suite Community
- Nikto
- Kali Linux
- Web browser

7.4 Methodology

1. Verify target connectivity.
2. Inspect HTTP responses.
3. Perform Nikto scanning.
4. Capture HTTP requests in Burp Suite.
5. Analyze responses.
6. Inspect HTTP headers.
7. Document observations.
8. Recommend mitigations.

7.5 Command

```
nikto -h http://testasp.vulnweb.com/
```

7.6 Screenshot Evidence

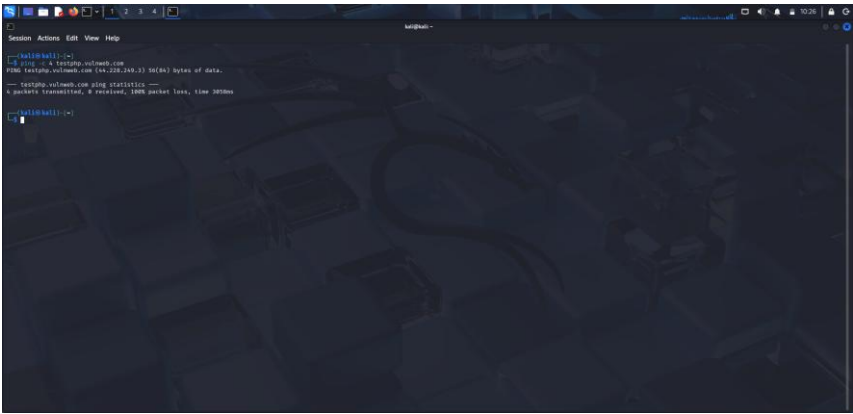


Figure 14: Target Connectivity Verification

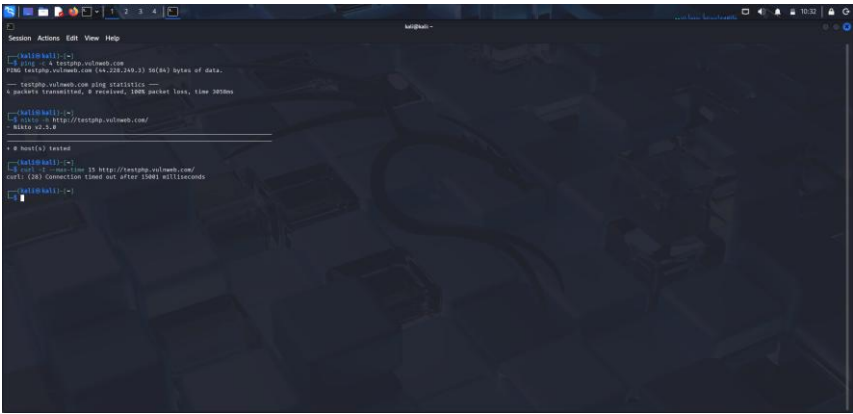


Figure 15: HTTP Response Verification Figure 16: TestASP Response

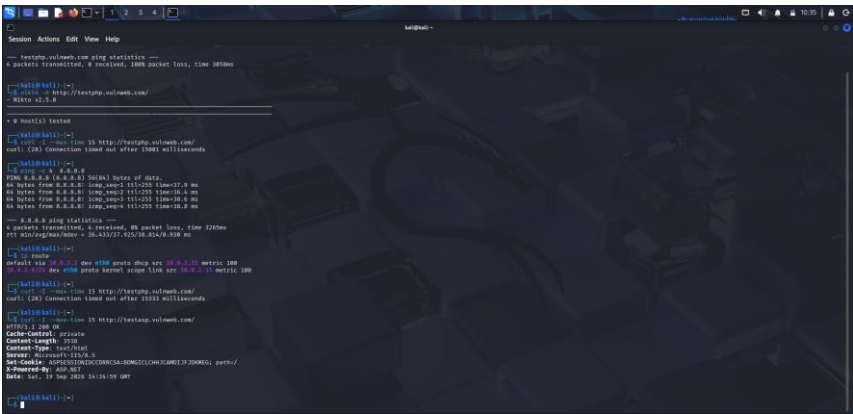


Figure 3: testasp HTTP Response

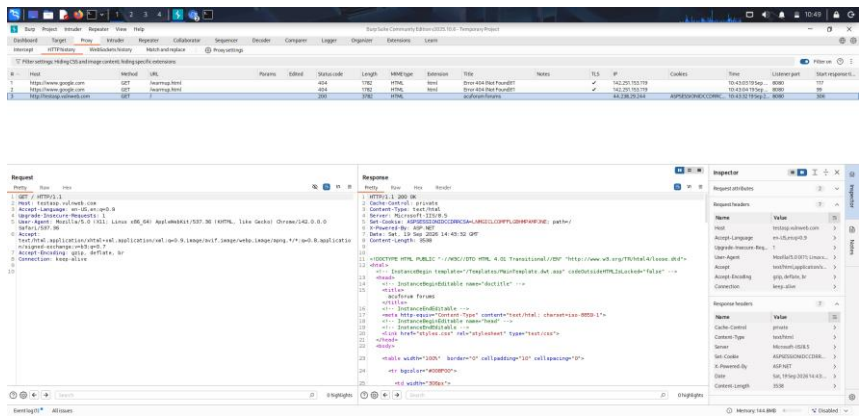


Figure 20: HTTP Response Headers

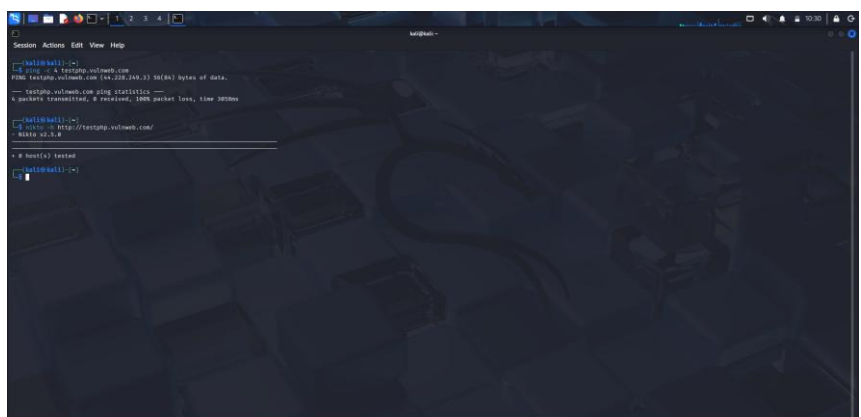


Figure 21: Nikto Assessment Evidence

7.7 Findings

Finding 1

Title: No Nikto Findings Reported

Evidence: Nikto scan

Description:

The Nikto scan was executed against the target web application. The displayed output shows 0 host(s) tested, and no specific vulnerability finding is reported in the captured output.

Impact:

No specific security impact can be determined from this screenshot because no Nikto vulnerability finding is shown.

Recommendation:

Perform the Nikto scan again with the correct target and ensure that the scan completes successfully before drawing vulnerability conclusions.

Finding 2

Title: Web Server and Technology Information Disclosure

Evidence: Burp Suite HTTP Response

Description:

The HTTP response captured in Burp Suite exposes server and technology information through response headers. The response includes Server: Microsoft-IIS/8.5 and X-Powered-By: ASP.NET.

Impact:

Exposing server and framework information may provide useful technology details to an attacker during reconnaissance.

Recommendation:

Configure the web server to minimize unnecessary technology disclosure by removing or restricting identifying response headers where possible.

7.9 Conclusion

This project provided practical experience with basic web vulnerability assessment and demonstrated how automated scanning and manual HTTP analysis can be used together.

5. ADVANCED LEVEL

OVERtheWIRE BANDIT LEVELS 0–20

LEVEL 0

Objective: Connect to the Bandit server and read the readme file.

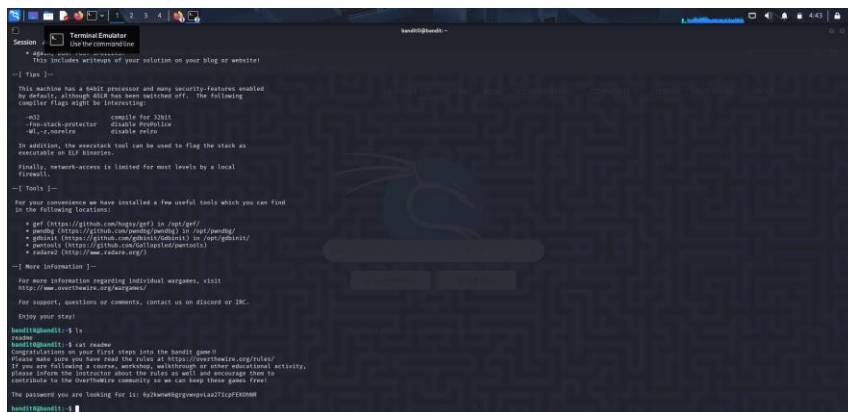
Commands:

```
ssh bandit0@bandit.labs.overthewire.org -p 2220
```

```
cat readme
```

Reasoning: The readme file contained the credential required to proceed.

Result: Next-level credential obtained.



```
Session | Terminal Emulator | Use the command window | bandit@bandit0-
+-----+-----+-----+
- [ Tips ] -
This machine has a 64bit processor and many security-features enabled
by default, although some have been matched off. The following
compiler flags might be interesting:
-wl2          compile for 32bit
-fno-stack-protector  disable StackSmash
-wl-r,rorelro  disable relro
In addition, the stackcheck tool can be used to flag the stack as
vulnerable on 64 bit processors.
Finally, network access is limited for most levels by a local
firewall.
- [ Tools ] -
For your convenience we have installed a few useful tools which you can find
in the following locations:
+ perl (https://github.com/hackplayers/perl) in /opt/perl/
+ perlsec (https://github.com/hackplayers/perlsec) in /opt/perlsec/
+ perlsec (https://github.com/hackplayers/perlsec) in /opt/perlsec/
+ perlsec (https://github.com/hackplayers/perlsec) in /opt/perlsec/
+ perlsec (https://github.com/hackplayers/perlsec) in /opt/perlsec/
- [ More Information ] -
For more information regarding individual weapons, visit
https://www.overthewire.org/weapons/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stage!
bandit@bandit0:~$ ls
bandit@bandit0:~$ cat readme
Congratulations on your first stage into the bandit game!
Please note that you have read the rules at https://overthewire.org/rules/
If you are planning a contest, workshop, challenge or other educational activity,
please inform the instructor about the rules as well and encourage them to
contribute to the Overthewire community so we can have these games free!
The password you are looking for is: h41m4n4g1ngr0wn1s477g1pr1n00m
bandit@bandit0:~$
```

Figure 22: Bandit Level 0 – Initial Access

LEVEL 1

Objective: Read the file named -.

```
ls -la
```

```
cat ./-
```

Reasoning: ./- explicitly identifies the filename rather than treating - as an option.

Result: Next-level credential obtained.

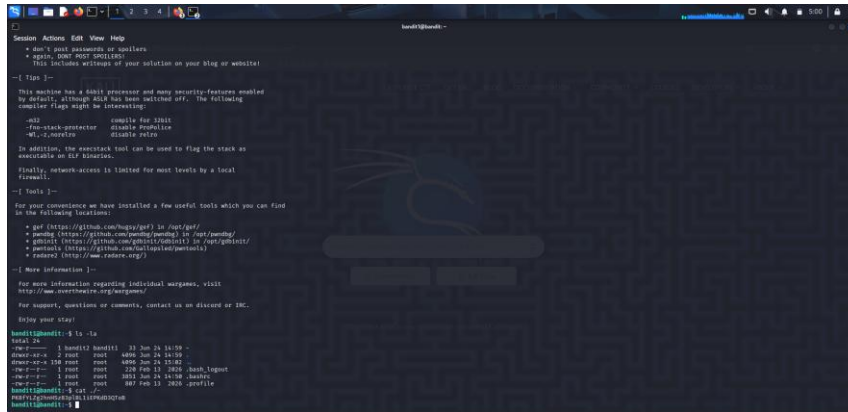


Figure 23: Bandit Level 1 – File Named –

LEVEL 2

Objective: Read the file containing spaces in its filename.

```
cat "./--spaces in this filename--"
```

Reasoning: Quotation marks preserve the complete filename as one argument.

Result: Next-level credential obtained.

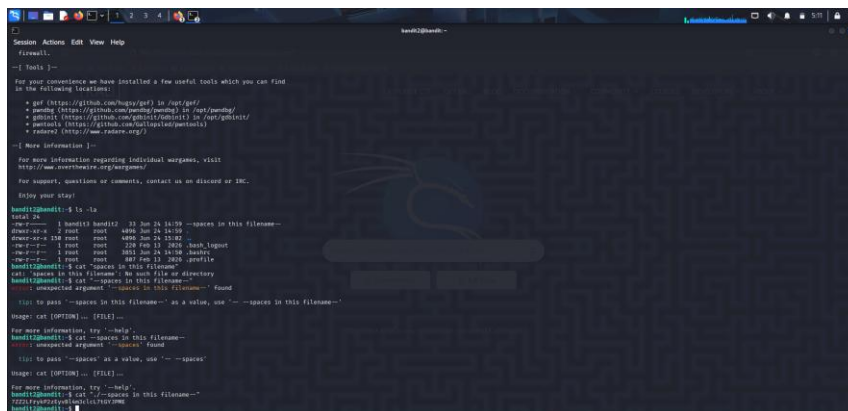


Figure 24: Bandit Level 2 – Filename with Spaces

LEVEL 3

Objective: Find and read the hidden file.

cd inher

ls -la

```
cat ./...Hiding-From-You
```

Reasoning: `ls -la` displays hidden files.

Result: Next-level credential obtained.

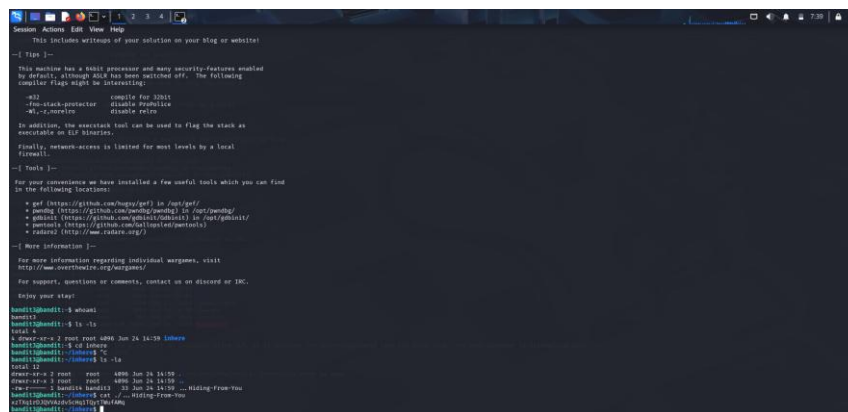


Figure 25: Bandit Level 3 – Hidden File Identification

LEVEL 4

Objective: Identify the human-readable file.

file ./*

cat ./file07

Reasoning: file identifies file types.

Result: Next-level credential obtained.

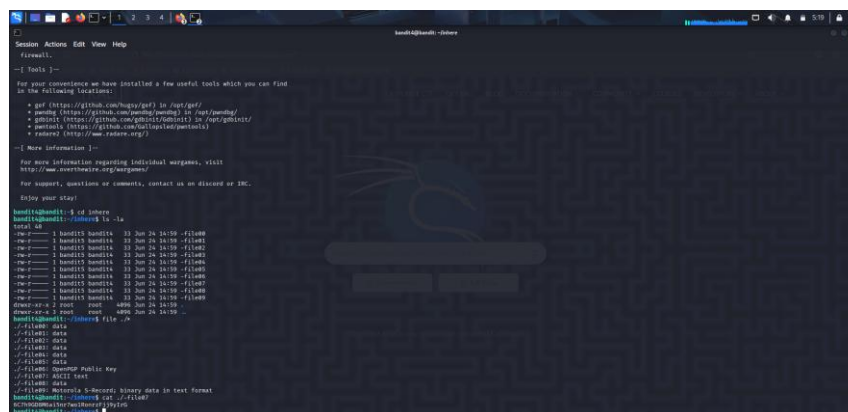


Figure 26: Bandit Level 4 – Human-Readable File Identification

LEVEL 5

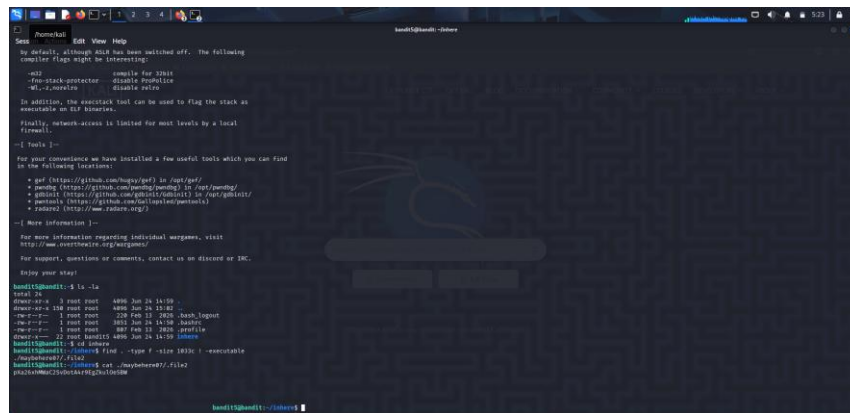
Objective: Locate a non-executable file matching the required size.

find . -type f -size 1033c ! -executable

Then:

cat ./maybehere07/.file2

Result: Next-level credential obtained.



```
bandit@bandit:~$ ls -la
total 28
drwxr-xr-x 3 root root 4096 Jun 28 14:28
drwxr-xr-x 138 root root 4096 Jun 28 14:28
-rw-r--r-- 1 root root 218 Jun 28 14:28 .bash_logout
-rw-r--r-- 1 root root 352 Jun 28 14:28 .bashrc
-rw-r--r-- 1 root root 592 Jun 28 14:28 .profile
-rw-r--r-- 2 root root 4096 Jun 28 14:28 .ssh
bandit@bandit:~$ cd /home
bandit@bandit:~/home$ find -type f -size 1431c ! -executable
./maybehere07/.file2
bandit@bandit:~/home$ cat ./maybehere07/.file2
p00t3sM0u325ub0tAr0g2u5u0n0W
```

Figure 27: Bandit Level 5 – File Attribute Search

LEVEL 6

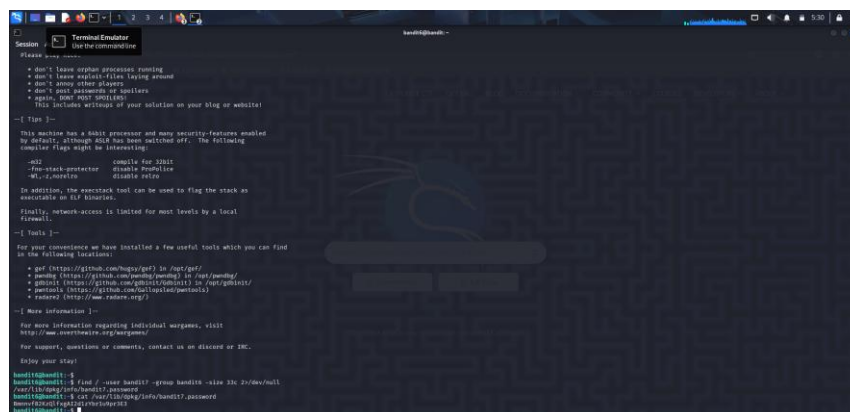
Objective: Search the filesystem using ownership and size conditions.

find / -user bandit7 -group bandit6 -size 33c 2>/dev/null

Resulting file:

/var/lib/dpkg/info/bandit7.password

Result: Credential obtained from the identified file.



```
bandit@bandit:~$ find / -user bandit7 -group bandit6 -size 33c 2>/dev/null
/var/lib/dpkg/info/bandit7.password
bandit@bandit:~$ cat /var/lib/dpkg/info/bandit7.password
b0unc3dP4xllRqg1n1r0b0t0p3R2
```

Figure 28: Bandit Level 6 – System-Wide File Search

LEVEL 7

Objective: Find the required word in data.txt.

grep millionth data.txt

```
Session Actions Edit View Help  
Please play nice:  


- don't leave orphan processes running
- don't leave sensitive files lying around
- don't annoy other players
- don't post passwords or spoilers
- again, DON'T ROOT HOSTS! :D

This includes warnings of your violation on your blog or website!  
  
-[ Tips ]-  
  
This machine has a debug processor and many security-features disabled by default, although ASLR has been switched off. The following compiler flags might be interesting:  


- mrel          compile for RHEL
- fno-stack-protector   disable StackSmash
- fcv=none       disable CXXASX

In addition, the rezerohack tool can be used to flip the stack as accessible on ELF binfiles.  
  
Finally, network-access is limited for most levels by a local firewall.  
  
-[ Tools ]-  
  
For your convenience we have installed a few useful tools which you can find at the following locations:  


- # get (https://github.com/hugsy/gpf) by /get/gpf/
- # patching (https://github.com/mymagnum/powmgw) by /get/powmgw/
- # exploit (http://github.com/0x0cc/cveexploit) by /get/cveexploit/
- # pentest (http://github.com/0x0cc/tools/pentests)
- # relay (http://www.relayr.org/)

  
-[ More information ]-  
  
For more information regarding individual webpages, visit http://www.mozartmachine.org/webpages.  
  
For support, questions or comments, contact us on discord or IRC.  
  
Enjoy your stay!  
  
loadelf@mandrill:~$  
loadelf@mandrill:~$ ls -la  
total 16  
loadelf@mandrill:~$ grep willlaunch data.txt  
loadelf@mandrill:~$ cat /dev/shm/fidmsnucg9zFqWkVtueub  
loadelf@mandrill:~$
```

LEVEL 8

```
sort data.txt | uniq -u
```

```

[+] Tips :-
  This machine has a shell processor and many security features enabled by default, although SUDO has been switched off. The following compiler flags might be interesting

  -m32      compile for 32bit
  -fno-stack-protector  disable StackProtect
  -U -mnoavx  disable avx

  In addition, the stackcheck tool can be used to flag the stack as overflows or ELF headers.

  Finally, network access is limited for most levels by a local firewall.

[+] Tools :-
  For your convenience we have installed a few useful tools which you can find in the following locations:

  # get (https://github.com/hackplayers/p0p) in /opt/p0p/
  # P0ping (https://github.com/p0pmelee/p0pmelee) in /opt/p0pmelee/
  # gmmcat (https://github.com/gmmcat/gmmcat) in /opt/gmmcat/
  # perlcat (https://github.com/Perlcat/Perlcat)
  # fudat2 (http://www.fudatw.org/)

  More Information :-
  For more information regarding individual wargames, visit
  http://www.overflow.wn/games/

  For support, questions or comments, contact us on discord or IRC.

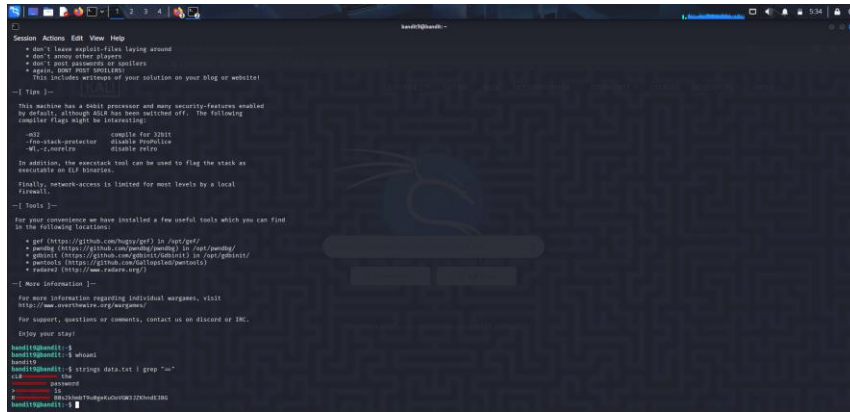
  Enjoy your level

root@kali:~# $ whoami
root@kali
root@kali:~# $ ls
data.txt
root@kali:~# $ cat data.txt | wc -l
1
root@kali:~# $ cat data.txt | xxd -l 1
00000000: 00000000 00000000 00000000 00000000
root@kali:~# $

```

LEVEL 9

```
strings data.txt | grep "=="
```



```
bandit10@bandit:~$ cat data.txt
The password is 0x00000000000000000000000000000000
```

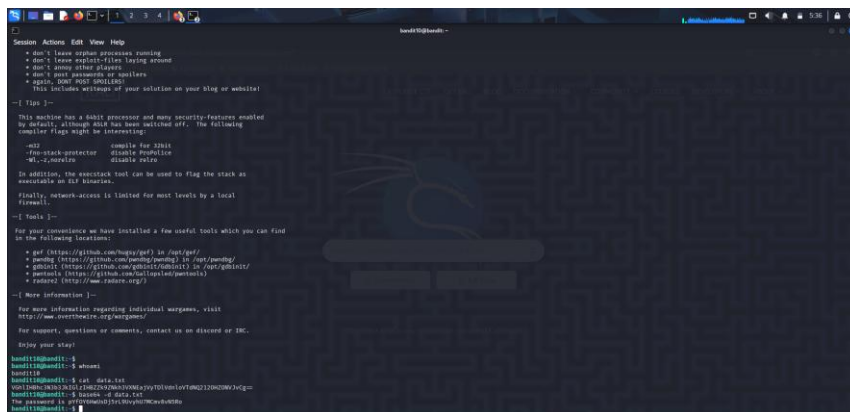
Figure 31: Bandit Level 9 – String Extraction

LEVEL 10

Objective: Decode Base64 data.

base64 -d data.txt

Result: Next-level credential obtained.



```
bandit10@bandit:~$ cat data.txt
The password is 0x00000000000000000000000000000000
```

Figure 32: Bandit Level 10 – Base64 Decoding

LEVEL 11

Objective: Decode ROT13 text.

cat data.txt | tr 'A-Za-z' 'N-ZA-Mn-za-m'

Result: Next-level credential obtained.

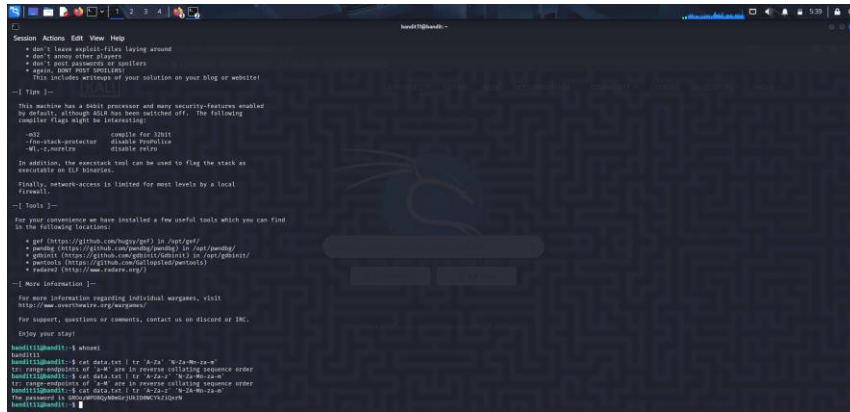


Figure 33: Bandit Level 11 – ROT13 Decoding

LEVEL 12

Objective: Extract and decompress multiple layers of data.

```
tar -xOf /tmp/data data5.bin | tar -xOf - data6.bin | bzip2 -d | tar -xOf - data8.bin | gzip -d
```

Reasoning: The pipeline progressively extracted and decompressed the nested data.

Result: Next-level credential obtained.

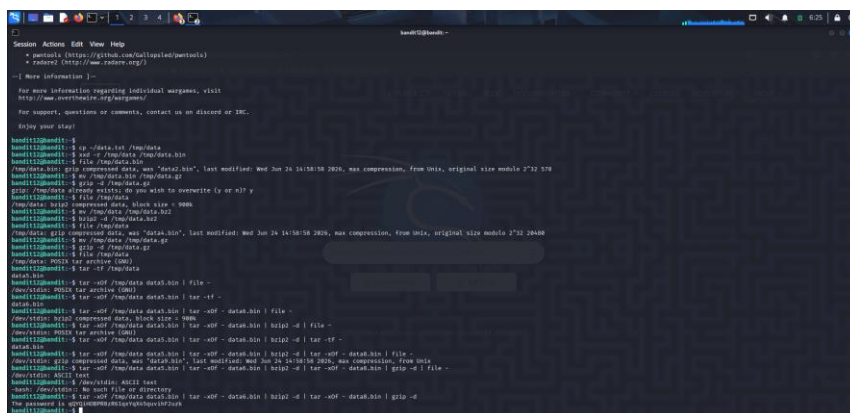


Figure 34: Bandit Level 12 – Multi-Layer Archive Extraction

LEVEL 13

Objective: Authenticate using the provided SSH private key.

```
chmod 600 ~/.sshkey.private
```

```
ssh -p 2220 -i ~/.sshkey.private bandit14@bandit.labs.overthewire.org
```

Result: Successfully connected to the next account.

```
Session Actions Edit View Help
Command "ddnsupdate" is in order to generate a random and hard to guess
directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc
filesystem so that users cannot browse an unencrypted file and subdirectory
with really horrible or smart names will be periodically deleted. The /tmp
directory is regularly wiped.

+ don't leave orphan processes running
+ don't leave exploit-files laying around
+ don't allow other players
+ don't post passwords or spoilers
+ again, DONT POST SPOILERS!
  (This includes writeups of your solution on your blog or website!)

-- Tip 3 --
This machine has a 64bit processor and many security-features enabled
by default, although mitM has been switched off. The following
console flags might be interesting:

-mitM          compile for 32bit
-jm-stack-protector  disable stack-protector
-Wl,-z,norelro  disable retp
In addition, the rstack tool can be used to flag the stack as
accessible on 32 bit systems.
Finally, network access is limited for most levels by a local
firewall.

-- Tools --
For your convenience we have installed a few useful tools which you can find
in the following locations:
+ get (https://github.com/hackplayers/get) in /opt/get/
+ pendu (https://github.com/pendu/pendu) in /opt/pendu/
+ phishit (https://github.com/phishit/phishit) in /opt/phishit/
+ pentest (https://github.com/pendu/pentest) in /opt/pentest/
+ pentest (https://www.pentest.org/)
-- More information --
For more information regarding individual weaknesses, visit
http://www.evotheatre.org/weaknesses/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stage!

bandit@bandit:~$
```

Figure 35: Bandit Level 13 – SSH Key Authentication

LEVEL 14

Objective: Send the credential to the local service.

nc localhost 30000

Result: Next-level credential obtained.

```
Session Actions Edit View Help
+ again, DONT POST SPOILERS!
  (This includes writeups of your solution on your blog or website!)

-- Tip 3 --
This machine has a 64bit processor and many security-features enabled
by default, although mitM has been switched off. The following
console flags might be interesting:

-mitM          compile for 32bit
-jm-stack-protector  disable stack-protector
-Wl,-z,norelro  disable retp
In addition, the rstack tool can be used to flag the stack as
accessible on 32 bit systems.
Finally, network access is limited for most levels by a local
firewall.

-- Tools --
For your convenience we have installed a few useful tools which you can find
in the following locations:
+ get (https://github.com/hackplayers/get) in /opt/get/
+ pendu (https://github.com/pendu/pendu) in /opt/pendu/
+ phishit (https://github.com/phishit/phishit) in /opt/phishit/
+ pentest (https://github.com/pendu/pentest) in /opt/pentest/
+ pentest (https://www.pentest.org/)
-- More information --
For more information regarding individual weaknesses, visit
http://www.evotheatre.org/weaknesses/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stage!

bandit@bandit:~$ ls -la
total 24
drwxr-xr-x 3 root root 4096 Jun 24 14:19 .
drwxr-xr-x 1 root root 4096 Jun 24 14:19 ..
-rw-r--r-- 1 root root 228 Feb 13 2024 .bash_logout
-rw-r--r-- 1 root root 228 Feb 13 2024 .bashrc
-rw-r--r-- 1 root root 228 Feb 13 2024 .profile
bandit@bandit:~$ cat /etc/bandit_pass/bandit14
www.evotheatre.org/weaknesses/
bandit@bandit:~$ nc localhost 30000
```

Figure 36: Bandit Level 14 – Localhost Network Connection

LEVEL 15

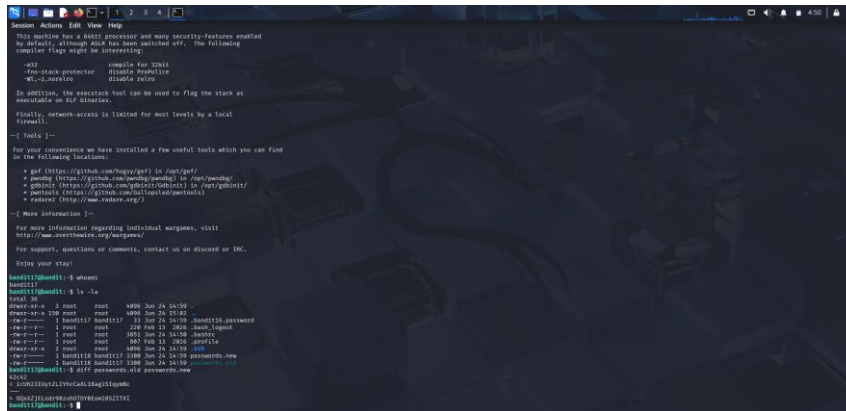
Objective: Connect to the SSL/TLS service.

openssl s_client -connect localhost:30001 -quiet

Result: Next-level credential obtained.

The line beginning with > represented the value in the newer file.

Result: Credential for the next level obtained.



```
bandit17@bandit:~$ diff passwords.old passwords.new
bandit17@bandit:~$
bandit17@bandit:~$ ls -la
total 36
drwxr-xr-x 3 root root 4096 Jun 24 14:30
drwxr-xr-x 100 root root 4096 Jun 24 14:32
-rw-r--r-- 1 bandit17 bandit17 33 Jun 24 14:30 bandit18.password
-rw-r--r-- 1 root root 228 Jun 13 24:24 band17.pwd
-rw-r--r-- 1 root root 202 Jun 20 24:08 bandit2
-rw-r--r-- 1 root root 887 Jun 13 24:24 g0tmi1k
-rw-r--r-x 1 bandit18 bandit18 1388 Jun 24 14:30 passwords.new
-rw-r--r-- 1 bandit18 bandit18 1388 Jun 24 14:30
bandit17@bandit:~$ diff passwords.old passwords.new
bandit17@bandit:~$
bandit17@bandit:~$ cat /etc/crontab/crontab
* * * * * root run-parts /etc/cron.d
bandit17@bandit:~$
```

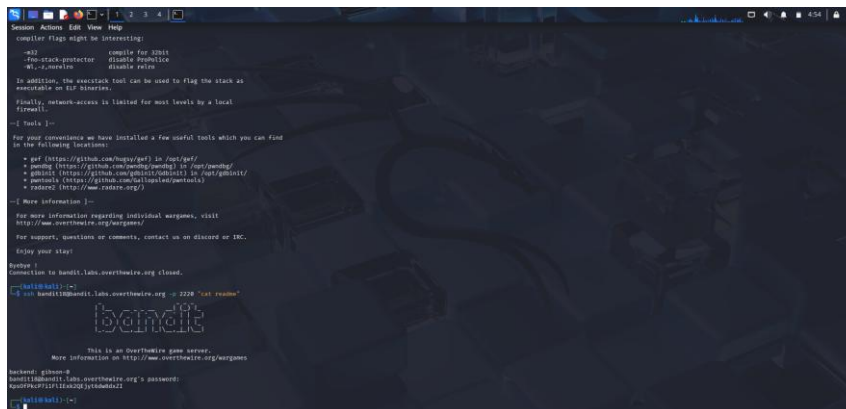
Figure 39: Bandit Level 17 – Password Difference Identification

LEVEL 18

Objective: Execute a remote command because the normal shell closes.

ssh bandit18@bandit.labs.overthewire.org -p 2220 "cat readme"

Result: Next-level credential obtained.



```
bandit18@bandit:~$ cat /etc/crontab/crontab
* * * * * root run-parts /etc/cron.d
bandit18@bandit:~$
bandit18@bandit:~$ ls -la
total 36
drwxr-xr-x 3 root root 4096 Jun 24 14:30
drwxr-xr-x 100 root root 4096 Jun 24 14:32
-rw-r--r-- 1 bandit17 bandit17 33 Jun 24 14:30 bandit18.password
-rw-r--r-- 1 root root 228 Jun 13 24:24 band17.pwd
-rw-r--r-- 1 root root 202 Jun 20 24:08 bandit2
-rw-r--r-- 1 root root 887 Jun 13 24:24 g0tmi1k
-rw-r--r-x 1 bandit18 bandit18 1388 Jun 24 14:30 passwords.new
-rw-r--r-- 1 bandit18 bandit18 1388 Jun 24 14:30
bandit17@bandit:~$ diff passwords.old passwords.new
bandit17@bandit:~$
bandit17@bandit:~$ cat /etc/crontab/crontab
* * * * * root run-parts /etc/cron.d
bandit17@bandit:~$
```

Figure 40: Bandit Level 18 – Remote Command Execution

LEVEL 19

Objective: Use the SUID executable to read the next credential.

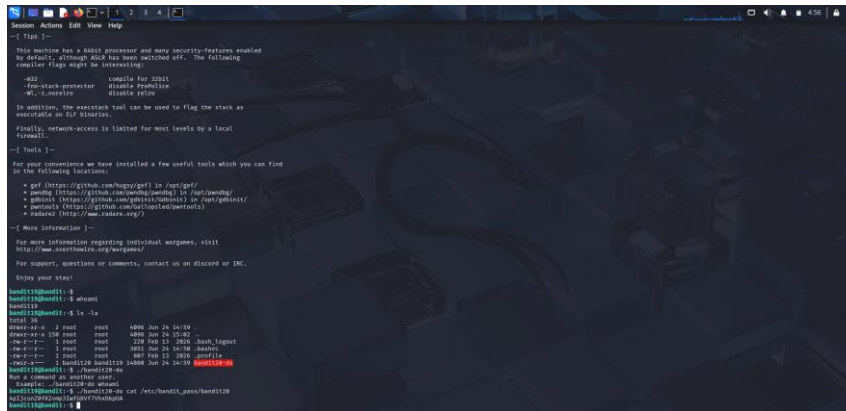
ls -la

./bandit20-do

Then:

`./bandit20-do cat /etc/bandit_pass/bandit20`

Result: Bandit Level 20 credential obtained.



```
Session Actions Edit View Help
-- Tip --
This machine has a 32bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:
-m32          compile for 32bit
-fno-stack-protector  disable ProPolice
-W,-fstackline  disable ASLR
In addition, the stackcheck tool can be used to flag the stack as
executable on ELF binaries.
Finally, network-access is limited for most levels by a local
firewall.
-- Tools --
For your convenience we have installed a few useful tools which you can find
in the following locations:
* gcc (https://github.com/hackplayers/gcc) in /opt/gcc/
* pwmap (https://github.com/pwmap/pwmap) in /usr/games/
* g0tmilk (https://github.com/g0tmilk/g0tmilk) in /usr/games/
* pwmap (https://github.com/pwmap/pwmap) in /usr/games/
* radare2 (http://www.radare.org/)
-- More information --
For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!
bandit19@bandit:~$ suid
bandit19@bandit:~$ ls -la
total 20
drwxr-xr-x 1 root root 4096 Jun 24 14:59
-rw-r--r-- 1 root root 4096 Jun 24 15:02
-rw-r--r-- 1 root root 228 Feb 11 2016 hash_input
-rw-r--r-- 1 root root 431 Jun 24 14:59 hash
-rw-r--r-- 1 root root 807 Feb 11 2016 profile
-rwxr-xr-x 1 bandit19 bandit19 1088 Jun 24 14:59 bandit20-do
bandit19@bandit:~$ ./bandit20-do
bandit19@bandit:~$ ./bandit20-do cat /etc/bandit_pass/bandit20
bandit20-do
bandit19@bandit:~$
```

Figure 41: Bandit Level 19 – SUID Privilege Execution

LEVEL 20

Objective: Use suconnect with a local listening service.

Terminal 1

`nc -l -p 12345`

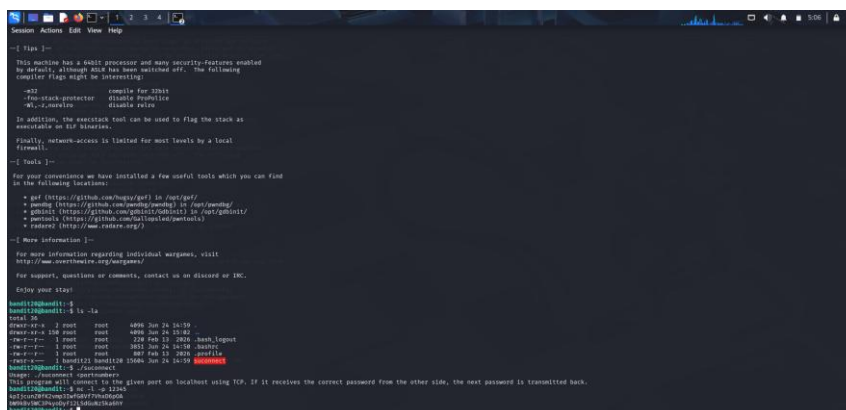
Terminal 2

`./suconnect 12345`

Reasoning: One terminal provided the listening service and the second terminal connected to that service through suconnect.

Result: The challenge was successfully completed.

Evidence 1:



```
Session Actions Edit View Help
-- Tip --
This machine has a 32bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:
-m32          compile for 32bit
-fno-stack-protector  disable ProPolice
-W,-fstackline  disable ASLR
In addition, the stackcheck tool can be used to flag the stack as
executable on ELF binaries.
Finally, network-access is limited for most levels by a local
firewall.
-- Tools --
For your convenience we have installed a few useful tools which you can find
in the following locations:
* gcc (https://github.com/hackplayers/gcc) in /opt/gcc/
* pwmap (https://github.com/pwmap/pwmap) in /usr/games/
* g0tmilk (https://github.com/g0tmilk/g0tmilk) in /usr/games/
* pwmap (https://github.com/pwmap/pwmap) in /usr/games/
* radare2 (http://www.radare.org/)
-- More information --
For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!
bandit20@bandit:~$ suid
bandit20@bandit:~$ ls -la
total 20
drwxr-xr-x 1 root root 4096 Jun 24 14:59
-rw-r--r-- 1 root root 4096 Jun 24 15:02
-rw-r--r-- 1 root root 228 Feb 11 2016 hash_input
-rw-r--r-- 1 root root 431 Jun 24 14:59 hash
-rw-r--r-- 1 root root 807 Feb 11 2016 profile
-rwxr-xr-x 1 bandit20 bandit20 1088 Jun 24 14:59 bandit21-do
bandit20@bandit:~$ ./bandit21-do
bandit20@bandit:~$ ./bandit21-do cat /etc/bandit_pass/bandit21
bandit21-do
bandit20@bandit:~$
```

Figure 42: Bandit Level 20 – Listener Terminal

Evidence 2:

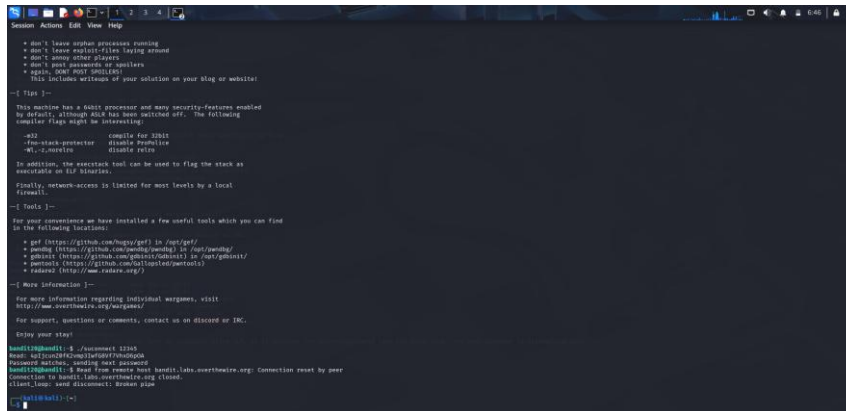


Figure 43: Bandit Level 20 – suconnect Terminal

6. CONSOLIDATED COMMANDS

Beginner

- `ping testphp.vulnweb.com`
- `nmap testphp.vulnweb.com`
- `nmap -sV testphp.vulnweb.com`
- `whatweb testphp.vulnweb.com`
- `nc testphp.vulnweb.com 80`
- `gobuster dir -u http://testasp.vulnweb.com/ -w /usr/share/wordlists/dirb/common.txt`
- `nikto -h testasp.vulnweb.com`

Bandit

- `ls -la`
- `cat readme`
- `cat ./-`
- `cat "./--spaces in this filename--"`
- `cd inhere`
- `file ./*`
- `find . -type f -size 1033c ! -executable`
- `find / -user bandit7 -group bandit6 -size 33c 2>/dev/null`
- `grep millionth data.txt`
- `sort data.txt | uniq -u`
- `strings data.txt | grep "=="`
- `base64 -d data.txt`
- `cat data.txt | tr 'A-Za-z' 'N-ZA-Mn-za-m'`
- `tar -xOf /tmp/data data5.bin | tar -xOf - data6.bin | bzip2 -d | tar -xOf - data8.bin | gzip -d`
- `chmod 600 ~/.sshkey.private`
- `ssh -p 2220 -i ~/.sshkey.private bandit14@bandit.labs.overthewire.org`
- `nc localhost 30000`
- `openssl s_client -connect localhost:30001 -quiet`
- `nmap -p 31000-32000 localhost`
- `openssl s_client -connect localhost:31790 -quiet`

- `diff passwords.old passwords.new`
- `ssh bandit18@bandit.labs.overthewire.org -p 2220 "cat readme"`
- `./bandit20-do cat /etc/bandit_pass/bandit20`
- `nc -l -p 12345`
- `./suconnect 12345`

7. SKILLS AND LEARNING OUTCOMES

Through the completed internship activities, I gained practical knowledge in:

- Network reconnaissance
- Port scanning
- Service enumeration
- Web technology identification
- Directory enumeration
- HTTP request analysis
- Web vulnerability assessment
- Linux command-line operations
- SSH authentication
- File permissions
- Encoding and decoding
- Compression/decompression
- TCP communication
- SSL/TLS connections
- SUID concepts
- Security documentation

The practical exercises also improved my ability to troubleshoot technical problems and approach cybersecurity challenges systematically.

8. OVERALL CONCLUSION

The CyberSNova internship provided practical exposure to multiple areas of cybersecurity.

I completed all four Beginner Level projects involving network reconnaissance, directory enumeration, login traffic analysis, and web vulnerability assessment.

I also completed the Advanced Level OverTheWire Bandit challenge from Level 0 through Level 20.

The projects helped me understand how cybersecurity professionals perform reconnaissance, enumeration, traffic analysis, vulnerability assessment, and Linux-based security tasks.

The internship also improved my familiarity with industry-relevant security tools and strengthened my understanding of responsible and authorized security testing.

9. FINAL SCREENSHOT CHECKLIST

Section	Required Evidence	Count
Project 1	Connectivity, Nmap Basic, Nmap Service, WhatWeb, Netcat, Web Ports	6
Project 2	Target Connectivity, HTTP Response, Gobuster Enumeration, Gobuster Results, Directory Verification	5
Project 3	HTTP POST URI, Login Form Data	2
Project 4	Connectivity, HTTP Response, testasp Response, Nikto, Burp Request, Burp Response, Headers, Nikto Assessment	8
Bandit	Levels 0–19; Level 20 – both terminals	21

Total: 42 screenshot evidences